

Transforming Neonatal Care: Smart H-IoT-based Cradle System for Real-time Monitoring and Personalized care

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Abstract

The birth of a newborn is a joyous occasion, yet it can be profoundly stressful for premature or sick infants requiring immediate admission to the Neonatal Intensive Care Unit (NICU). Despite advancements in neonatal care from traditional, in-person consultations to remote monitoring, privacy concerns about health data remain a critical challenge.

The study proposes a novel H-IoT (Healthcare - Internet of Things) Smart Infant cradle architecture featuring a user-friendly data acquisition framework equipped with optimized sensor integration. This system aims to empower parents with real-time monitoring, precise data insights, and adequate access to infants' health information. This ensures seamless data sharing between parents and healthcare professionals, thus abridging the existing gaps. The system's core consists of a NodeMCU controller board integrated with a set of sensors- sound, temperature (DHT 11), and moisture for comprehensive monitoring of the infant's environment and physiological parameters. Real-time data is displayed on a cradle-integrated tablet and simultaneously transmitted to a central server (Ubidots and MQTT Server) for further analysis. Phototherapy treatment is optimized through an adaptive mechanism that adjusts temperature based on real-time infant body temperature fluctuations. Furthermore, an external camera enables remote video monitoring and instant notifications via a designated mobile application, providing real-time alerts and actionable insights. The proposed system not only alleviates the burden on medical staff but enhances neonatal care through a secure data-driven approach with seamless data integration and management making it easier for parents to balance their professional and family responsibilities.

Keywords: H-IoT, NodeMCU, NICU, DHT11

1. Introduction

Changing Neonatal Care: Real-time monitoring and individualized care via an intelligent, Internet of Things-based cradle system

This innovative framework transforms neonatal care by providing real-time monitoring and personalized care. Equipped with sensors to track the infant's vitals, temperature, and movement, it empowers medical professionals with continuous data for early intervention. Advanced machine learning algorithms can detect anomalies alerting caregivers to potential threats. Parents can remotely access their baby's health data through an interactive user interface, which would foster peace of mind and a stronger bond. A secured cloud storage feature enhances the overall monitoring process by ensuring data privacy and accessibility. The wearable sensors offer comprehensive health monitoring. Furthermore, the data analytics feature can help identify potential patterns and provide customized health alerts, thus optimizing care plans.

2. Literature survey

Transforming neonatal care through a Smart H-IoT (Healthcare-Internet of Things) based cradle system for real-time monitoring and personalized care is a groundbreaking approach that addresses the challenges faced by modern parents, especially working mothers. The Smart Baby Cradle System leverages IoT technology to provide continuous monitoring of an infant's vital parameters, such as temperature, wetness, and crying, using sensors integrated into the cradle. This system, designed with components like Arduino UNO microcontrollers, moisture sensors, sound sensors, and DHT11 sensors, ensures that parents can monitor their baby's condition remotely via applications like Blynk, which displays real-time data and allows control over features like music and fans [1]. The automated nursery concept further enhances this by integrating various sensors and a microprocessor to monitor the infant's actions and environmental conditions, providing real-time updates and alerts to parents, thus conserving their time and energy [3]. Additionally, the IoT-based smart cradle systems are equipped with features such as automated rocking mechanisms, adjustable inclines, and soothing lullabies, which can be controlled remotely through mobile applications, offering a personalized sleeping experience for the baby [7]. These systems also incorporate advanced technologies like Raspberry Pi modules, gas sensors, sound sensors, and camera surveillance to capture every movement and ensure the baby's safety from anywhere in the world [9]. The integration of cloud-based platforms allows for the storage and analysis of sensor data, enabling parents to monitor vital signs, receive alerts, and control the cradle's functionalities through web or mobile apps [5]. The inclusion of machine learning models to detect emotions and cry patterns makes the system feature rich, reducing the need for manual intervention[8]. The feasibility and scalability of such systems are also addressed,

ensuring data privacy, security, and long-term viability, making them a trusted resource for parents [2]. Overall, the Smart H-IoT based cradle systems represent a significant advancement in neonatal care, offering enhanced safety, comfort, and convenience for both babies and caregivers, and transforming the way infant care is managed in today's fast-paced world [4] [10].

3. *Methodology*

The development of an Android-enabled smart cradle necessitates a structured flow with a stage-wise approach based on designed parameters and operational requirements. This section talks about the core components and methodical approach utilized in the Smart Infant Cradle utilizing H-IoT technology [11].

1.1 System Requirements and Core Components

The initial phase delves into the thorough identification of requirements based on the structural and system design of the proposed innovation. The system requirements are consolidated based on extensive literature research drawing inferences from the existing frameworks along with consultation and feedback from an established panel of medical professionals specializing in this field across Public and Private Healthcare settings [12]. This also encompasses the viewpoint of the caregivers and operational environment, keeping in view the need to understand the problem from each one's perspective.

The portable design structure stems from the choice of suitable microcontroller board for integration with a sensor suite including - temperature, load cell, sound, motion, and humidity detection for monitoring of vital parameters, further complemented by sophisticated data collection and analysis protocols with establishment of remote monitoring functionalities. This also features an integrated LED Monitor that serves as a patient monitor analog. The LED Monitor comprises two modes - Wellness (Monitor) and Cozy (Normal) Mode. The wellness mode is responsible for tracking vital signs - heart rate, oxygen saturation, blood pressure, and ECG waveforms. This sets a standardized template for continuous monitoring with real time updates directed to parents via Mobile App through the central cloud server being the designated health data repository, acting as a common point of data exchange.

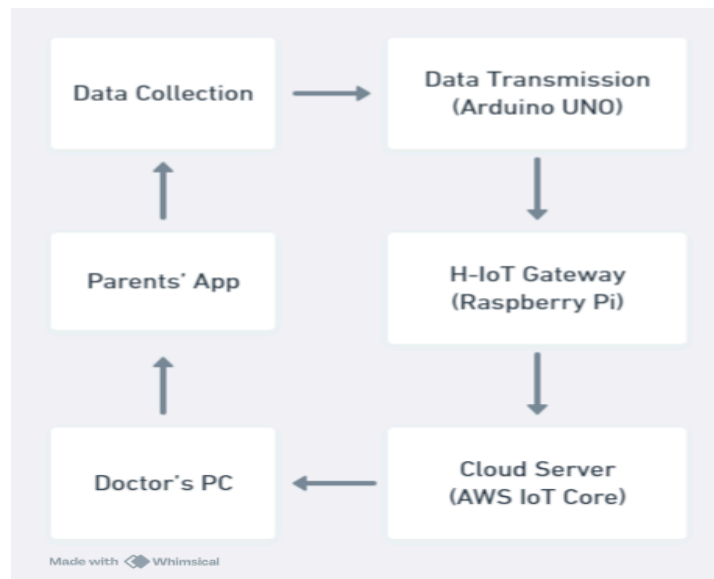
1.2 Data Acquisition Workflow :

The data transfer from the microcontroller board through sensors employs a stepwise approach for data preprocessing and analysis. The acquired sensor data is channeled to a microcontroller (ATMEGA 6250) or IoT gateway for preliminary processing and aggregation[3]. This processed data is subsequently transmitted to a cloud-based platform

for comprehensive storage. Wireless communication protocols like cellular networks, Bluetooth, and Wi-Fi facilitate data transmission for visualization and analysis.

Following data preprocessing and analysis, the proposed system intends to extract meaningful insights from the infant's well-being viewpoint by advanced analytics and machine learning algorithms [13]. The data access will be provided through two channels- a central cloud based server system for healthcare professionals through which data is rooted and shared via a mobile application solely developed to share real time alerts for guardian's access [14]. This enables remote infant monitoring, including real-time alerts for elevated sound levels and video calls in critical situations.

Figure 1 provides a visual representation of the Smart H-IoT Cradle's operational mechanism.



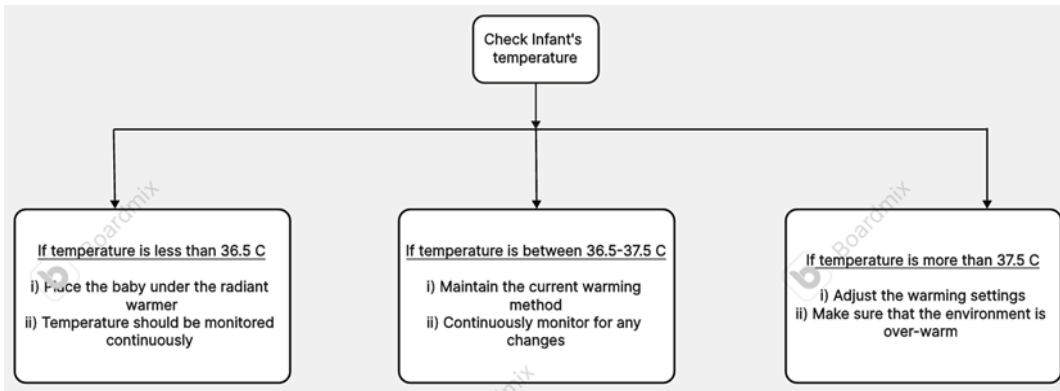
(Fig 1. Data analysis mechanism)

1.3 The customized framework within the core components

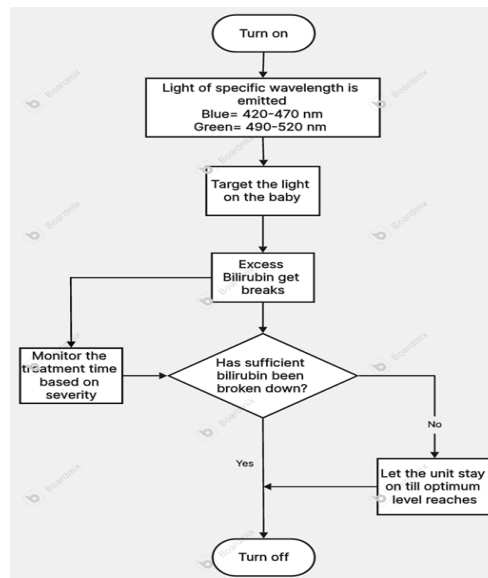
Neonatal Intensive Care Units (NICUs) commonly employ warming units to maintain optimal infant temperature. The heat loss experienced by newborns is managed by warming units that serve as essential support to simulate a thermoregulated environment as per requirement. There are two kinds of warmers - manual and automatic warmers. The proposed smart cradle incorporates an automatic warming unit [15]. Once the infant is placed, the sensors detect environmental and infant temperatures, and further, adjust the heating elements accordingly to achieve optimized temperature. Figure 2 represents the mechanism of the warming unit.

Phototherapy is an exclusive treatment for neonatal jaundice, a condition that is characterized by elevated bilirubin levels in the infant's blood. The general range of bilirubin is from 0.3 to 1.0 mg/dL within the first 24 hours of life, with levels exceeding 13 mg/dL necessitating immediate intervention. The treatment involves exposure of the infant to specific wavelengths of light (approximately 400-500 nm) to facilitate bilirubin breakdown. In our smart cradle, the unit is activated to deliver the required light spectrum, and treatment duration is monitored based on bilirubin levels. Once optimal bilirubin levels are achieved, the phototherapy unit is deactivated. The innovative design incorporates a detachable phototherapy unit in our case.

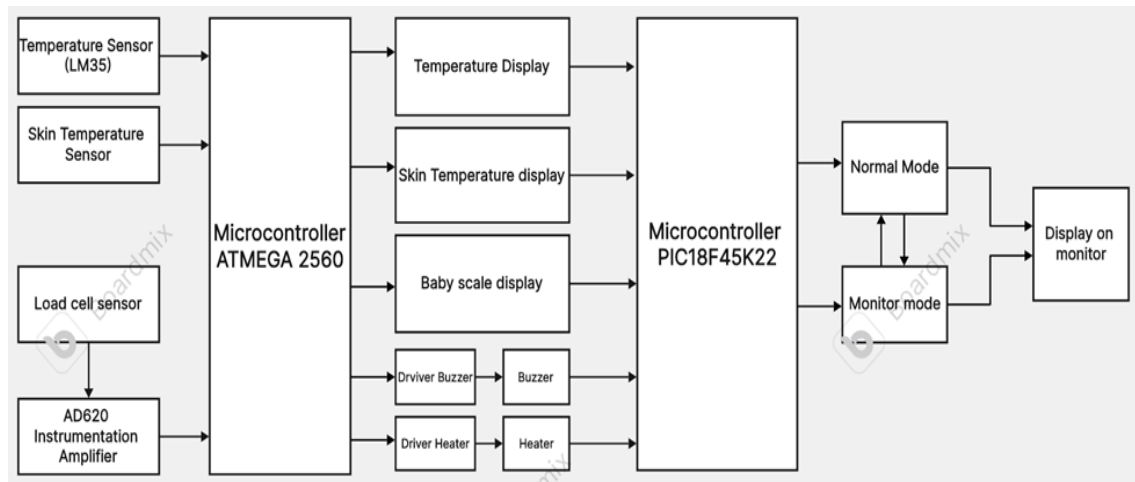
Figure 3 and 4 represents the mechanism of the phototherapy unit and the overall mechanism of the working of the Smart H-IoT Infant Cradle, respectively.



(Fig 2. Mechanism of the warming unit)



(Fig 3. Mechanism of Phototherapy unit)



(Fig 4. Mechanism of Smart H-IoT Infant Cradle)

4. Code and Algorithm

4.1 Interactive UI creating algorithm

It defines the entry point of the flutter application and configures the app theme and home screen.

```

// lib/vital_signs.dart
class VitalSigns {
  final double heartRate;
  final double temperature;
  final double respiratoryRate;
  VitalSigns({
    required this.heartRate,
    required this.temperature,
    required this.respiratoryRate,
  });
}

// lib/main.dart
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
import 'vital_signs_provider.dart';
void main() {
  runApp(MyApp());
}

```

```

}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return ChangeNotifierProvider(
      create: (context) => VitalSignsProvider(),
      child: MaterialApp(
        title: 'Infant Vitals App',
        theme: ThemeData(
          primarySwatch: Colors.blue,
        ),
        home: VitalSignsScreen(),
      ),
    );
  }
}

class VitalSignsScreen extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    final vitalSigns = Provider.of<VitalSignsProvider>(context).vitalSigns;
    return Scaffold(
      appBar: AppBar(
        title: Text('Infant Vital Signs'),
      ),
      body: Padding(
        padding: const EdgeInsets.all(16.0),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text('Heart Rate: ${vitalSigns.heartRate} bpm', style: TextStyle(fontSize: 24)),
            Text('Temperature: ${vitalSigns.temperature} °C', style: TextStyle(fontSize: 24)),
            Text('Respiratory Rate: ${vitalSigns.respiratoryRate} breaths/min', style:
TextStyle(fontSize: 24)),
            SizedBox(height: 20),
            ElevatedButton(
              onPressed: () {
                Provider.of<VitalSignsProvider>(context, listen: false).updateVitalSigns(130, 37.0, 32);
              },
              child: Text('Update Vital Signs'),
            ),
          ],
        ),
      ),
    );
  }
}

```

```

],
),
),
);
}
}

```

4.2 Dependencies & configuration settings algorithm

This defines the dependencies, assets and other configuration settings for a project.

Dependencies:

Flutter:

sdk: flutter

provider: ^6.0.0

4.3 Vitals display algorithm

It is used for clarity and maintainability, it's generally used to keep the project file(dart file) well-structured and readable.

// lib/vital_signs_provider.dart

```
import 'package:flutter/material.dart';
```

```
import 'vital_signs.dart';
```

```
class VitalSignsProvider with ChangeNotifier {
```

```
VitalSigns _vitalSigns = VitalSigns(heartRate: 120, temperature: 36.5, respiratoryRate: 30);
```

```
VitalSigns get vitalSigns => _vitalSigns;
```

```
void updateVitalSigns(double heartRate, double temperature, double respiratoryRate) {
```

```
_vitalSigns = VitalSigns(heartRate: heartRate, temperature: temperature, respiratoryRate: respiratoryRate);
```

```
notifyListeners();
```

```
}
```

```
}
```

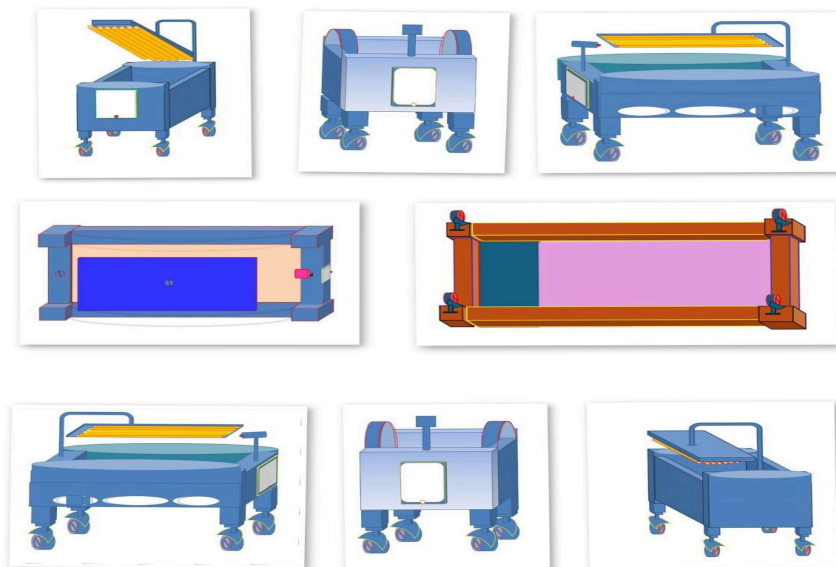
5. *Results and Discussion*

The development of the Smart H-IoT Cradle has led to significant advancements in neonatal care, driven by its innovative design and integrated functionalities. The framework has been enhanced through its portability, with the incorporation of warming and phototherapy units, along with a robust analytics framework that ensures data privacy and security. The cradle design prioritizes portability while maintaining functionality. The innovative structure allows for easy relocation within healthcare settings, ensuring immediate access to infants in need. The compact warming unit effectively maintains

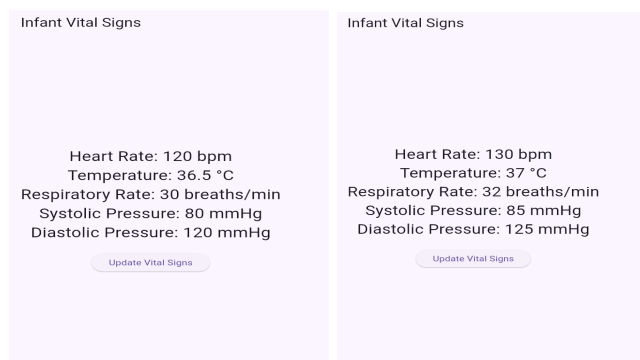
optimal temperature levels, while the phototherapy unit addresses jaundice making it a versatile instrument for caregivers.

A key feature of the Smart Cradle is its robust analytics system, which safeguards sensitive health data. The system facilitates compartmentalized data sharing, enhancing both security and accessibility. This dual-channel approach fosters collaboration between caregivers and healthcare providers. This is quite essential for enhancing secured data management and actionable insights. Stringent security measures, including data encryption and compliance with healthcare regulations, ensure the privacy of sensitive health information. The analytics framework not only improves the quality of care for infants but also strengthens the partnership between caregivers and healthcare providers.

Figure 5 and 6 represents different views of Cradle and output of vitals display app, respectively.



(Fig 5. Different views of Cradle)



(Fig 6. Output of vitals display app)

6. Conclusion :

The Smart H-IoT cradle system not only intends to revolutionize and optimize neonatal care through a portable system framework but also enhances the data management aspect and sets a precedent for future innovations in this field. By bridging the gap between technology and healthcare, this system exemplifies the potential of the Internet of Things (IoT) to enhance patient outcomes, improve caregiver efficiency, and ultimately contribute to the well-being of the vulnerable newborn population. As the healthcare sector continues to embrace technological advancements, the insights gleaned from this study will serve as a foundation for future research and development in smart healthcare solutions at the Home-Hospital Interface for continuous data monitoring with robust analytics architecture that would enhance personalized care.

7. References

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